Inorganic Cume

October 8, 1999

If the following reactions are considered as acid-base reactions, which reagent or reagent component would be the acid and which would be the base? Explain your reasoning in each case:

1. $I^- + I_2 \rightarrow I_3^-$
2. $PF_5 + NaF \rightarrow NaPF_6$
3. $Fe + 5CO \rightarrow Fe(CO)_5$

Ka_1 for H_2TeO_3 is $1x10^{-3}$.

4. Estimate the value of Ka_2 for H_2TeO_3. Explain the reasoning behind your estimate.

Aqueous Fe^{+3} solutions are seldom neutral.

5. Write the chemical equation that accounts for the non-neutral properties of Fe^{+3} solutions.

6. Which would be more acidic, 1 M Fe^{+2} or 1 M Fe^{+3}? Why?

Which of the following is the stronger acid?

7. H_3BO_3 or H_3PO_3? Why?
8. $HBrO_2$ or $HBrO_3$? Why?
9. HF or HCl? Why?

Which is the better Lewis acid?

10. BMe_3 or B(isopropyl)$_3$? Why?
11. BCl_3 or BF_3? Why?

Predict which way the following reactions would be favored to go (right or left)

12. $AlI_3 + 3NaF \leftrightarrow AlF_3 + 3NaI$

13. $CuI_2 + 2CuF \leftrightarrow CuF_2 + 2CuI$

14. $CH_3HgF + HSO_3^- \leftrightarrow CH_3HgSO_3^- + HF$

$R_2SBX_3 + R_2O \leftrightarrow R_2OBX_3 + R_2S$ goes to the right when X=F and to the left when X=H.

15. Explain this observation about these reactions.

The structures of the following adducts gave the N-B distances listed.

	N-B distance
CH_3CNBF_3	1.63A
H_3NBF_3	1.60A
$(CH_3)NBF_3$	1.58A

16. Explain the trend of the distances given.

17. Predict the direction of the trend in the B-F distances for the above, and explain your reasoning.